

In [1]: `import pandas as pd`

In [2]: `rd = pd.read_excel(r'D:\fsds\projects\datasets\Rawdata.xlsx')`
`rd`

Out[2]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [3]: `id(rd)`

Out[3]: 2205038421648

In [4]: `rd.columns`

Out[4]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')

In [5]: `rd.shape`

Out[5]: (6, 6)

In [6]: `rd.head()`

Out[6]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

In [7]: `rd.tail()`

Out[7]:

	Name	Domain	Age	Location	Salary	Exp
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [8]: `rd.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain       6 non-null      object
2   Age         4 non-null      object
3   Location    4 non-null      object
4   Salary      6 non-null      object
5   Exp         5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [9]: `rd`

Out[9]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [10]: `rd.isnull()`

Out[10]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [11]: rd.isna()
```

```
Out[11]:
```

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [12]: rd.isna().sum()
```

```
Out[12]: Name      0
Domain      0
Age         2
Location    2
Salary      0
Exp         1
dtype: int64
```

```
In [13]: rd['Name']
```

```
Out[13]: 0      Mike
1      Teddy^
2      Uma#r
3      Jane
4      Uttam*
5      Kim
Name: Name, dtype: object
```

```
In [14]: rd['Name'] = rd['Name'].str.replace(r'\W', '', regex=True)
```

```
In [15]: rd['Name']
```

```
Out[15]: 0      Mike
1      Teddy
2      Umar
3      Jane
4      Uttam
5      Kim
Name: Name, dtype: object
```

```
In [16]: rd
```

Out[16]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [17]: `rd['Domain'] = rd['Domain'].str.replace(r'\W', '', regex=True)`
`rd['Domain']`

Out[17]:

0	Datascience
1	Testing
2	Dataanalyst
3	Analytics
4	Statistics
5	NLP

Name: Domain, dtype: object

In [18]: `rd`

Out[18]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [19]: `rd['Age'] = rd['Age'].str.replace(r'\W', '', regex=True) # replace(r'\W', '', re`
`rd`

Out[19]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34years	Mumbai	5^00#0	2+
1	Teddy	Testing	45yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [20]: `rd['Age'] = rd['Age'].str.extract(r'(\d+)') # extract(r'(\d+)') will extract dig`
`rd`

Out[20]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5^00#0	2+
1	Teddy	Testing	45	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67	NaN	30000-	5+ year
5	Kim	NLP	55	Delhi	6000^\$0	10+

```
In [21]: rd['Salary'] = rd['Salary'].str.replace(r'\W', '', regex=True)
rd
```

Out[21]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2+
1	Teddy	Testing	45	Bangalore	10000	<3
2	Umar	Dataanalyst	NaN	NaN	15000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5+ year
5	Kim	NLP	55	Delhi	60000	10+

```
In [22]: rd['Exp'] = rd['Exp'].str.replace(r'\W', '', regex=True)
rd['Exp'] = rd['Exp'].str.extract(r'(\d+)')
rd
```

Out[22]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [23]: cd = rd.copy()
cd
```

Out[23]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

EDA Methods from now on

In [24]:

```
cd
```

Out[24]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [25]:

```
cd.isnull().sum()
```

Out[25]:

```
Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

In [26]:

```
cd['Age']
```

Out[26]:

```
0      34
1      45
2      NaN
3      NaN
4      67
5      55
Name: Age, dtype: object
```

In [27]:

```
import numpy as np
```

In [28]:

```
cd['Age'] = cd['Age'].fillna(np.mean(pd.to_numeric(cd['Age'])))
cd['Age']
```

```
Out[28]: 0      34
         1      45
         2    50.25
         3    50.25
         4      67
         5      55
         Name: Age, dtype: object
```

```
In [29]: cd['Exp'] = cd['Exp'].fillna(np.mean(pd.to_numeric(cd['Exp'])))
         cd['Exp']
```

```
Out[29]: 0      2
         1      3
         2      4
         3    4.8
         4      5
         5     10
         Name: Exp, dtype: object
```

```
In [30]: cd
```

```
Out[30]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	NaN	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [31]: cd['Location'] = cd['Location'].fillna(cd['Location'].mode()[0])
         cd['Location']
```

```
Out[31]: 0      Mumbai
         1    Bangalore
         2    Bangalore
         3    Hyderbad
         4    Bangalore
         5      Delhi
         Name: Location, dtype: object
```

```
In [32]: cd
```

Out[32]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [33]: `cd.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     object
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [34]: `cd['Age'] = cd['Age'].astype(int)`

In [35]: `cd.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int32
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: int32(1), object(5)
memory usage: 396.0+ bytes
```

In [36]: `cd['Salary'] = cd['Salary'].astype(int)`
`cd.info()`


```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int32
3   Location    6 non-null     object
4   Salary      6 non-null     int32
5   Exp         6 non-null     object
dtypes: int32(2), object(4)
memory usage: 372.0+ bytes
```

```
In [37]: cd['Exp'] = cd['Exp'].astype(int)
cd.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int32
3   Location    6 non-null     object
4   Salary      6 non-null     int32
5   Exp         6 non-null     int32
dtypes: int32(3), object(3)
memory usage: 348.0+ bytes
```

```
In [38]: cd['Name'] = cd['Name'].astype('category')
cd['Domain'] = cd['Domain'].astype('category')
cd['Location'] = cd['Location'].astype('category')
cd.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null     category
1   Domain      6 non-null     category
2   Age         6 non-null     int32
3   Location    6 non-null     category
4   Salary      6 non-null     int32
5   Exp         6 non-null     int32
dtypes: category(3), int32(3)
memory usage: 866.0 bytes
```

```
In [39]: cd.to_csv('cd.csv') # to convert dataframe to csv file
```

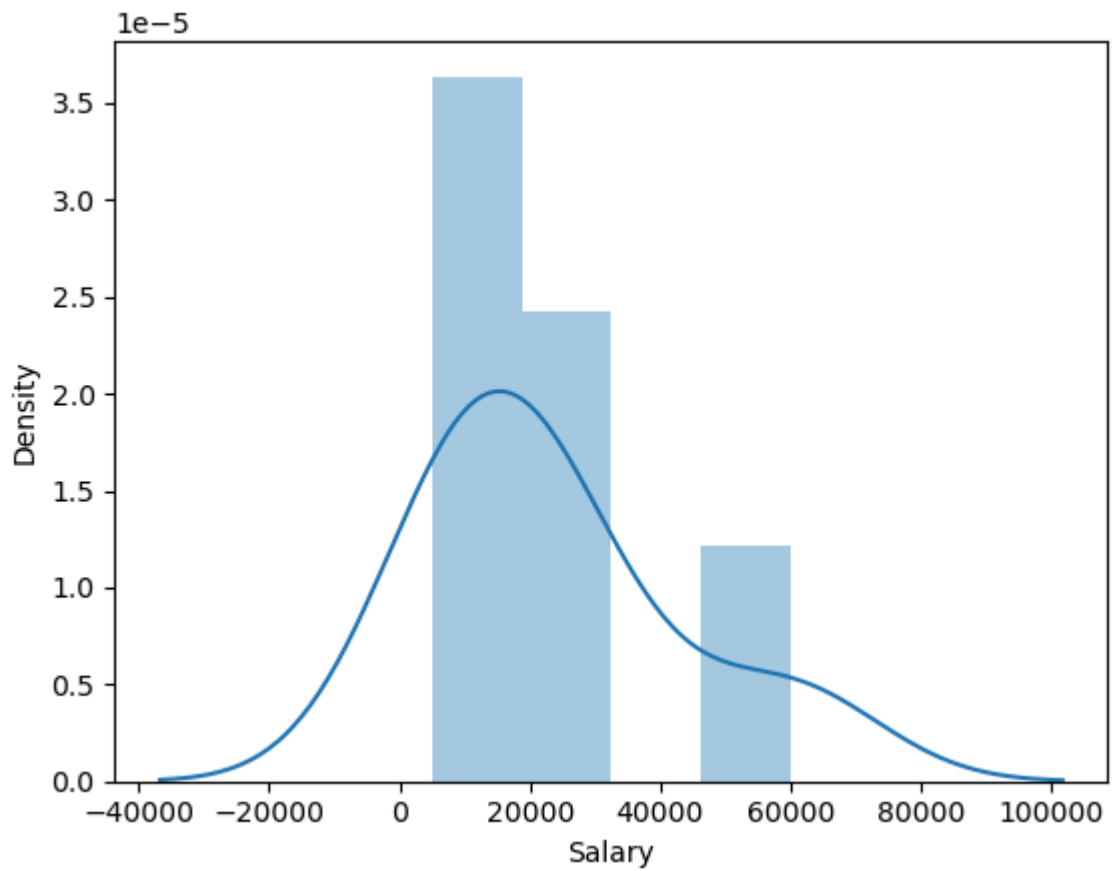
```
In [40]: import os
os.getcwd()
```

```
Out[40]: 'd:\\fsds\\projects'
```

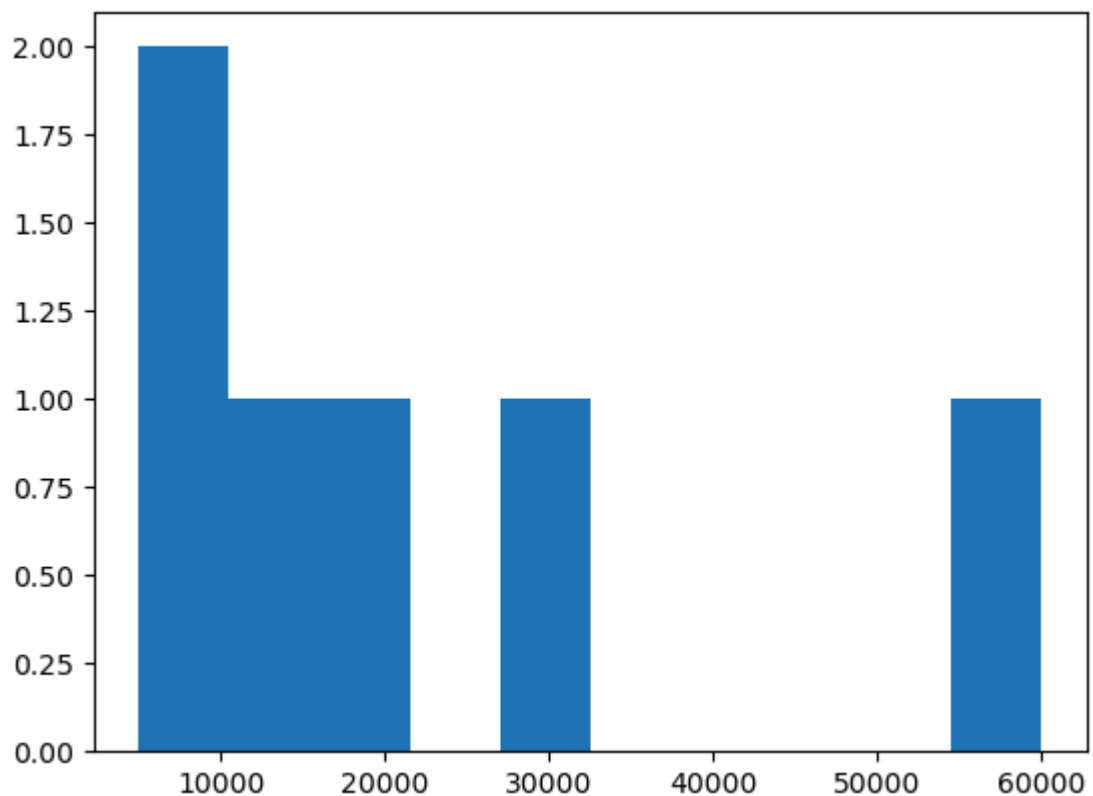
```
In [41]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [42]: import warnings  
warnings.filterwarnings('ignore')
```

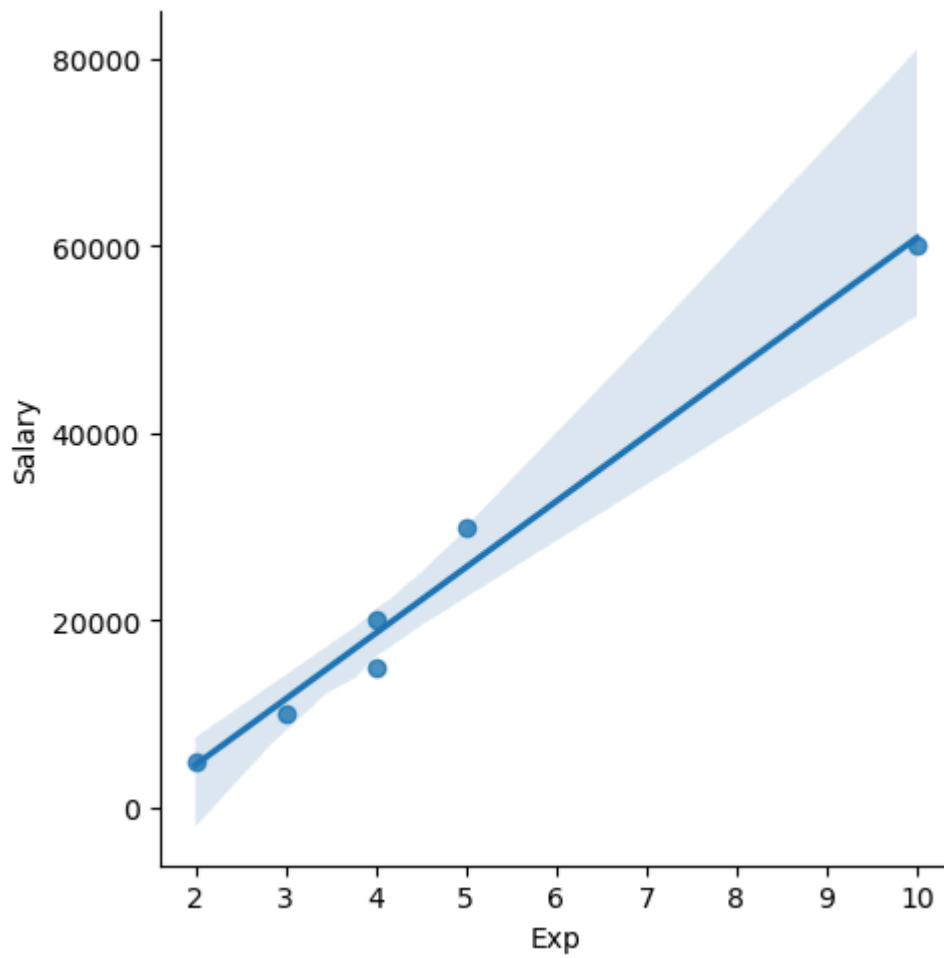
```
In [43]: vis1 = sns.distplot(cd['Salary'])
```



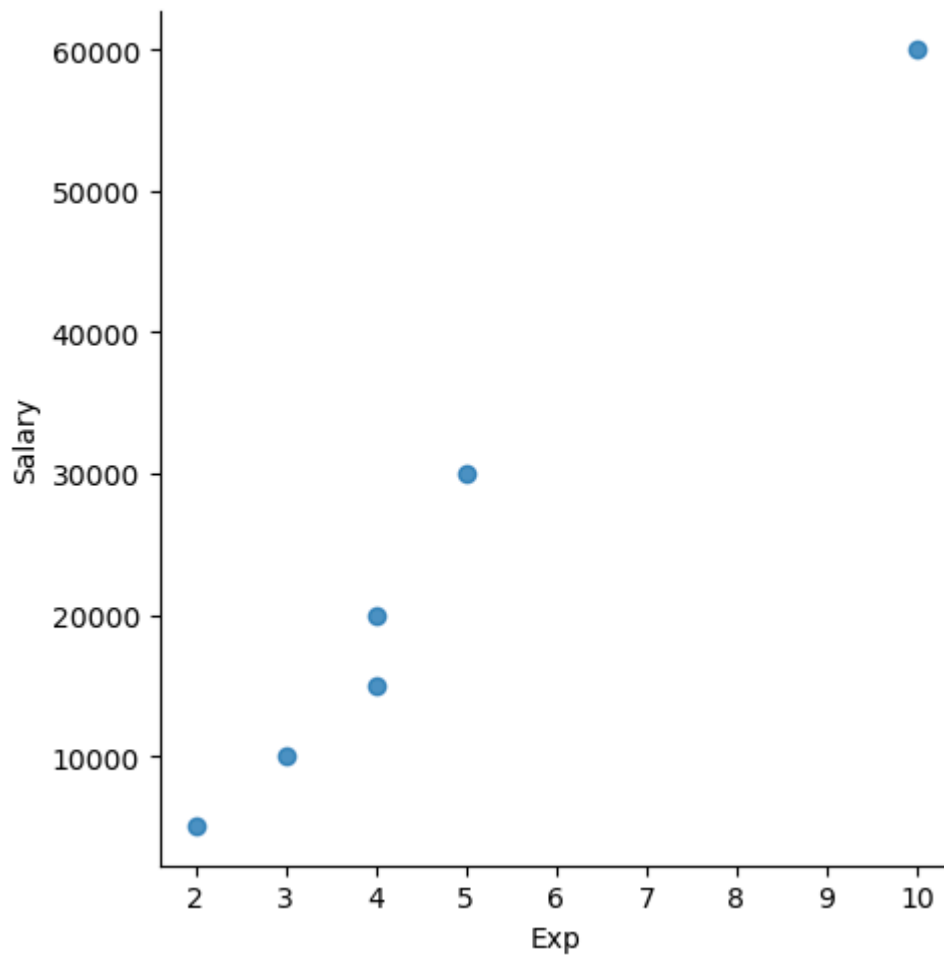
```
In [44]: vis2 = plt.hist(cd['Salary'])
```



```
In [46]: vis3 = sns.lmplot(data=cd, x = 'Exp', y='Salary')
```



```
In [48]: vis4 = sns.lmplot(data = cd, x='Exp', y='Salary', fit_reg=False)
```



```
In [49]: cd[:]
```

```
Out[49]:
```

	Name	Domain	Age	Location	Salary	Exp
--	------	--------	-----	----------	--------	-----

0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [50]: cd[0:6:3]
```

```
Out[50]:
```

	Name	Domain	Age	Location	Salary	Exp
--	------	--------	-----	----------	--------	-----

0	Mike	Datascience	34	Mumbai	5000	2
3	Jane	Analytics	50	Hyderabad	20000	4

```
In [51]: cd[:, :-1]
```

```
Out[51]:
```

	Name	Domain	Age	Location	Salary	Exp
5	Kim	NLP	55	Delhi	60000	10
4	Uttam	Statistics	67	Bangalore	30000	5
3	Jane	Analytics	50	Hyderabad	20000	4
2	Umar	Dataanalyst	50	Bangalore	15000	4
1	Teddy	Testing	45	Bangalore	10000	3
0	Mike	Datascience	34	Mumbai	5000	2

```
In [52]: cd.columns
```

```
Out[52]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [53]: x_iv=cd[['Name', 'Domain', 'Location', 'Age', 'Exp']]
y_dv=cd['Salary']
print(x_iv.shape, y_dv.shape)
```

```
(6, 5) (6,)
```

```
In [54]: y_dv
```

```
Out[54]: 0      5000
1     10000
2     15000
3     20000
4     30000
5     60000
Name: Salary, dtype: int32
```

```
In [55]: x_iv
```

```
Out[55]:
```

	Name	Domain	Location	Age	Exp
0	Mike	Datascience	Mumbai	34	2
1	Teddy	Testing	Bangalore	45	3
2	Umar	Dataanalyst	Bangalore	50	4
3	Jane	Analytics	Hyderabad	50	4
4	Uttam	Statistics	Bangalore	67	5
5	Kim	NLP	Delhi	55	10

```
In [57]: imputation= pd.get_dummies(cd)
imputation
```

Out[57]:

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar
0	34	5000	2	False	False	True	False	False
1	45	10000	3	False	False	False	True	False
2	50	15000	4	False	False	False	False	True
3	50	20000	4	True	False	False	False	False
4	67	30000	5	False	False	False	False	False
5	55	60000	10	False	True	False	False	False

In [58]: `imputation = imputation.astype(int)`
`imputation`

Out[58]:

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar
0	34	5000	2	0	0	1	0	0
1	45	10000	3	0	0	0	1	0
2	50	15000	4	0	0	0	0	1
3	50	20000	4	1	0	0	0	0
4	67	30000	5	0	0	0	0	0
5	55	60000	10	0	1	0	0	0

In [59]: `cd`

Out[59]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [60]: `imputation`

Out[60]:

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar
0	34	5000	2	0	0	1	0	0
1	45	10000	3	0	0	0	1	0
2	50	15000	4	0	0	0	0	1
3	50	20000	4	1	0	0	0	0
4	67	30000	5	0	0	0	0	0
5	55	60000	10	0	1	0	0	0



In []: